ENGINEER
a world of
DIFFERENCE

WALTER SCOTT, JR.
COLLEGE OF ENGINEERING
COLORADO STATE UNIVERSITY

Engineering Manufacturing Education Center (EMEC)

ABOUT THE SPACE:

The EMEC is not just a machine shop. It is the place where mechanical engineers (MECH) take the ideas they care most about and move them from their minds into the real world. It is unique among undergraduate programs that CSU Mechanical Engineering requires a comprehensive shop class where students are instructed in the safe and effective use of machine tools. This experience engages students. They become better designers and problem solvers and the experience equips them to rapidly innovate in the areas that interest them most.

In the EMEC, students learn more than just manufacturing. They learn courage, collaboration across disciplines, how to explore the physical design space, and how to think like three-dimensional problem solvers —honing their design intuition. Our students are recognized in industry for their ability to design, manufacture, test, and deploy solutions quickly, both as interns and as early career professionals.

Through this 2nd year lab course, students gain practical skills and experience by completing three major hands-on projects that connect classroom theory to real-world practice:

- Clock Project - Students machine a clock composed of an aluminum base, a brass pencil holder and nut, a steel pen holder, and an acrylic face. This project introduces manual machining processes and the interpretation of engineering drawings, using tools like the vertical mill and horizontal lathe.

- Screwdriver Project – Students produce two screwdrivers: a plastic injection-molded Phillips head screwdriver and an aluminum cast, drop-forged flathead screwdriver. These projects expose students to industry-standard processes such as casting and injection molding while deepening their manual machining skills.

- Coaster Project – Students design and manufacture an aluminum coaster cut with aerospace-inspired patterns and engraved with an image of their choosing. This project emphasizes CNC machining, one of the most prevalent technologies in modern manufacturing. Students also complete an oral exam that may qualify them for a HAAS Basic Mill Operator Certification.